

Genome-Wide Identification and Phylogenetic Analysis of the ERF Gene Family in Melon

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Abstract Members of the ERF transcription factor family have significant functions in regulating gene expression in response to biotic and abiotic stresses. In melon (*Cucumis melo* L.), however, few ERF genes have been studied. In this study, we performed a comprehensive computational analysis of 119 ERF family genes and identified 136 ERF superfamily genes in melon. We present a complete overview of ERF family gene in melon, including the phylogeny, gene structures, and putative conserved motifs of melon ERF family proteins, as well as a comparative analysis between ERF subfamily genes in cucumber. The AP2/ERF gene superfamily in melon can be classified into three families, ERF (ethylene-responsive factor), RAV (related to ABI3/VP1), and AP2 (APETALA2), which comprise 119, 4, and 13 members, respectively. The ERF family in melon was divided into 11 groups, designated I to XI, as in tomato. Our motif analysis indicated that most conserved motifs outside of the AP2/ERF domain are distributed between different clades in the phylogenetic tree. Most ERF family genes have no introns. Based on the observation that 11 of these groups existed in melon, we concluded that the chief functional diversification in the

ERF family predated the divergence between monocots and dicots. In our analysis of expression patterns, ERF subfamily genes were distributed widely throughout melon plant tissues and most of the ERF subfamily genes were detected in all seven types of tissues. Some ERF subfamily genes show expression patterns characteristic of tissue-specific expression, implying their probability of performing various roles in growth and development in melon.

Keywords Melon · ERF family · Phylogeny · Gene function

Introduction

The AP2/ERF superfamily is a large group of plant-specific transcription factors (Riechmann and others 2000), the members of which bear a characteristic AP2/ERF DNA-binding domain, which consists of 60–70 highly conserved amino acids (Nakano and others 2006). Based on sequence similarities and the numbers of AP2/ERF domains, this superfamily comprises 3 families. The AP2 family of proteins contains 2 repeat AP2/ERF domains, ERF proteins contain a single AP2/ERF domain, and RAV members harbor a B3 domain, in addition to a single AP2/ERF domain (Magnani and others 2004). The ERF family is further classified into 2 subfamilies: ERF and DREB (Sakuma and others 2002). In DREB proteins, amino acid 14 of the AP2/ERF domain is usually occupied by valine (important), but residue 19 in the DREB DNA-binding domain is glutamic acid (insignificant) (Sakuma and others 2002). Most proteins in the ERF subfamily have an alanine at position 14 and aspartic acid at position 19 in the ERF DNA-binding domain (Sharoni and others 2011). The AP2 domain was first isolated and characterized as a highly

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conserved repeated motif in *Arabidopsis thaliana*. AP2 family proteins have critical functions in flower development (Jofuku and others 1994; Irish 1998; Klucher and others 1996; Schmid and others 2003; Aida and others 2004; Nilsson and others 2006; Guillaumota and others 2008; Zhuang and others 2008). Ethylene-responsive element-binding proteins, which contain the ERF domain, were first identified as a conserved motif in 4 DNA-binding proteins in *Nicotiana tabacum*—EREBP1, 2, 3, and 4, which are currently named ERF1, 2, 3, and 4. This element binds specifically to a GCC box, a DNA sequence that mediates the ethylene-responsive transcription of genes (Ohme-Takagi and Shinshi 1995). With regard to the RAV family, RAV1 and 2 were discovered as full-length cDNAs for proteins that contain an AP2/ERF domain and a B3 domain in *Arabidopsis* (Kagaya and others 1999).

Melon (*Cucumis melo* L.) is a significant eudicot diploid plant species due to its specific properties and economic importance (Garcia-Mas and others 2012). It belongs to the cucurbitaceae family, which comprises 130 genera, including approximately 800 species that are found primarily in temperate, subtropical and tropical regions (Jeffrey 2005; Jeffrey and De Wilde 2006), and other valuable species, such as cucumber (*Cucumis sativus* L.), watermelon [*Citrullus lanatus* (Thunb.) Matsum. & Nakai], and squash (*Cucurbita spp.*) (Garcia-Mas and others 2012). Melon is an important fleshy fruit with regard to fresh consumption—its unique flavor renders it a favorite fruit for desserts and salads in the Americas, Europe, and the Middle East (Clepet and others 2011). In 2009, 26 million tons of melons were produced worldwide (<http://faostat.fao.org>). Melon is particularly important in Mediterranean and East Asian countries, where hybrid varieties have significant and climbing economic value (Garcia-Mas and others 2012). Melon is an attractive model for studying biological properties, including sex determination (Boualem and others 2008; Martin and others 2009), vascular biology (Haritatos and others 1996; Gomez and others 2005), and fruit ripening (Pech and others 2008), exhibiting extreme diversity in fruit traits. Melon species are highly polymorphic: melon fruits vary markedly in fruit shape, size, rind form, firmness, flesh color, ripening behavior, and flavor (Nunez-Palenius and others 2008). In addition, melon fruits also have significant variations in ripening physiology and are categorized as climacteric or nonclimacteric, based on their ripening-related respiration rate and ethylene evolution profiles (Giovannoni 2007). Melon fruit is a potential alternative model to tomato (*Solanum lycopersicum* L.) for studies of fleshy fruit ripening (Ezura and Owino 2008; Giovannoni 2007).

The International Cucurbit Genomics Initiative (ICuGI) database (available at <http://www.icugi.org> [verified Apr 8, 2012]) comprises approximately 8,144,030 expressed

sequence tags (ESTs) and 611,438 unigenes, nearly one-third of which were derived from fruit libraries (Portnoy and others 2011). In addition, MELONOMICS (<https://melonomics.net>) is available for genomic database which includes CM_protein_v3.5 (verified May 4, 2011), melon_genome_pseudomolecules_v3.5 (verified Jun 21, 2012), CM_assembly_scaffold_and_unscaffold_contigs_v3.5 (verified May 4, 2011). Completion of the genomic sequence of melon has allowed us to perform exhaustive phylogenetic comparisons with *Arabidopsis thaliana*, tomato, cucumber, and other plant species. The genomic sequence of melon is an invaluable tool that can be used to determine the genomic structure, gene function, and evolution of species.

AP2/ERF proteins have significant functions in the transcriptional regulation of many processes in growth and development, as well as responses to environmental stimuli. Recently, several ERF subfamily genes in plants have been cloned and examined with regard to function. Three AP2/ERF family genes—*Cme-DREB1*, *Cme-ERF1*, and *Cme-ERF2*—were cloned from a melon cDNA library by yeast one-hybrid screen with *Cme-ACS2* DRE/CRT cis-acting elements GCCGAC sequence within this gene promoter in melon. The former belongs to the DREB subfamily, and the latter two genes are in the ERF subfamily (Mizuno and others 2006). By functional analysis, *Cme-DREB1* activates the transcription of a downstream reporter gene through the GCCGAC sequence, and *CME-ERF1* and *CME-ERF2* are not involved in *Cme-ACS2* expression. Genes for the *Brassica napus* transcription factors BraERF1, 2, and 3 were identified, and their expression was analyzed (Zhuang and others 2008). Ei-Sharkawy and others (2009) cloned 7 ERF genes from peach and examined their expression patterns. The AP2/ERF transcription factor *SIAP2a* was characterized in tomato and determined to be a negative regulator of fruit ripening (Chung and others 2010). *SIERF6* was cloned in tomato and has a significant function in fruit ripening and carotene accumulation (Lee and others 2012).

After the release of the melon genome (Cucurbitaceae Genome Initiative 2012), 136 genes were postulated to encode proteins that contain the AP2/ERF domain, 87.5 % (119 genes) of which belong to the ERF family. All ERF subfamily members have been examined, despite the likelihood that they mediate many processes in plants. Much experimental work will be required to determine the specific function of each gene. Based on phylogenetic analyses, it is apparent that a large family of transcription factors comprises subgroups of genes that are closely related (Kranz and others 1998; Părenicova and others 2003; Toledo-Ortiz and others 2003; Reyes and others 2004; Tian and others 2004). Expression pattern analysis of each transcription factor in the ERF subfamily should be performed, taking into account functional redundancy

(Nakano and others 2006). To this end, an assessment of the structural relationships between all melon ERF family proteins can guide the prediction of their functions. Moreover, the availability of the cucumber (*Cucumis sativus* L.) genomes will allow us to compare the ERF subfamily between melon, which will be useful for studying function—particularly fleshy fruit ripening (Mizuno and others 2006; Chung and others 2010; Lee and others 2012; Sharma and others 2010; Licausi and others 2010; Yin and others 2010).

In this study, we attempted to generate a comprehensive overview of the ERF family in melon. Genes in the AP2/ERF superfamily in melon were surveyed, resulting in the identification of 136 genes in this superfamily, including 119 genes in the ERF family. Phylogenetic analyses of ERF family genes were performed, as were structural analyses of exons and introns and protein motifs. We also compared the phylogeny of cucumber (same family in Cucurbitaceae) ERF subfamilies.

The resulting classification of groups (ERF family genes), identification of putative functional motifs (ERF family genes), and expression pattern analysis (ERF subfamily genes) will be useful in studies on the function of each gene in the ERF families.

Materials and Methods

Identification of ERF Family Genes in Melon Genome

The AP2/ERF domain of the melon ethylene response factor sequence (GenBank number AB125975) was used as a query sequence for BLASTp (Altschul and others 1990) searches of AP2/ERF superfamily genes in the melon genome. CM_protein_v3.5 (<https://melonomics.net/files/Genome>), which was predicted as a protein sequence of the melon genome by MELONOMICS (<https://melonomics.net>), was used as a search database for local BLASTp. BioEdit 7.0.9 (Hall 1999) was used to analyze homologs of AP2/ERF superfamily genes in local BLASTp. Default parameters of the BLASTp program were E values $\leq 10^{-3}$.

We also searched the database using amino acid sequences, such as RAV, AP2, and the soloist gene At4g13040, of the AP2/ERF domain of members of the *Arabidopsis* AP2/ERF family as the query to confirm that the search was exhaustive (Zhuang and others 2009). In addition, the same sequences that were obtained with MELONOMICS by hidden Markov model (HMM) analysis with Pfam number PF00847, which contains a typical AP2/ERF domain (Wang and others 2010). The HMM profile of the AP2/ERF superfamily was extracted from Pfam.

To confirm the putative AP2/ERF superfamily genes, the amino acid sequences were searched for the AP2/ERF domain using the Simple Modular Architecture Research Tool (SMART) (Schultz and others 1998; Letunic and others 2009).

Multiple Sequence Alignment and Phylogenetic and MEME Motif Analysis

Multiple alignment analysis was performed with ClustalW (<http://bioinformatics.ubc.ca/resources/tools/clustalw>) (Chenna and others 2003). DNAMAN, version 7.0, was also used as a secondary tool to align sequences and verify the results. Another multiple sequence alignment was performed to compare the evolutionary relationships between melon and cucumber ERF subfamily members using DNAMAN, version 7.0, which yielded melon ERF and 125 cucumber ERF (available at <http://plantfdb.cbi.edu.cn/family.php?sp=Csg&fam=ERF>) subfamily protein members.

Phylogenetic trees were constructed using the neighbor-joining (NJ) method (Saitou and Nei 1987) with default parameters. Bootstrap analysis was performed using 1,000 replicates, and the phylogenetic trees were drawn with MEGA 5.10 (<http://www.megasoftware.net/mega.html>) (Tamura and others 2007). Another phylogenetic tree was constructed by the maximum parsimony (MP) method using MEGA 5.10 (Tamura and others 2007) to confirm the reliability of the NJ method. A combination of phylogenetic tree with melon ERF and cucumber ERF subfamily protein sequences was applied with MEGA 5.10 by the NJ method and a bootstrap of 1,000 replicates.

MEME Suite, version 4.8.1 (Bailey and Elkan 1995; Bailey and others 2006; Bailey and others 2009) (<http://meme.nbcr.net/meme/cgi-bin/meme.cgi>), was used with the following parameters: optimum width of 60–80 amino acids, any number of repetitions of a motif, and maximum number of motifs set to 5. The significance of the resulting motifs was checked in the NCBI and SMART databases (Sharma and others 2010).

Expression Analysis of Melon ERF Subfamily Genes in Melon

The expression characteristics of ERF subfamily genes were detected using expressed sequence tags (ESTs). We used the nucleotide sequences of ERF subfamily genes which are available from MELONOMICS (<https://melonomics.net>) as probing sequences for the proteins in melons. The performing of BLASTn in the NCBI EST database (<http://www.ncbi.nlm.nih.gov/blast.cgi>) was completed. As a result, a total of 23,152 EST sequences were obtained. Screening the sequences that had a

Table 1 Summary of AP2/ERF superfamily genes in melon and their comparison with cucumber, tomato, soybean, and *Arabidopsis*

Group	<i>Cucumis melon</i>	<i>Cucumis cucurbit</i> ^a	<i>Solanum lycopersicon</i> ^b	<i>Glycine max</i> ^c	<i>Arabidopsis thaliana</i> ^d
AP2	13	20	16	26	18
ERF	119	103	93	120	122
CBF/DREB	55	42	25	36	57
ERF	64	61	60	62	58
Others	–	–	–	–	7
Incomplete AP2/ERF domains	–	–	8	22	–
RAV	4	4	3	2	6
Soloist	–	4	–	–	1
Total	136	131	112	148	147

^a Hu and Liu (2011)^b Sharma and others (2010)^c Zhang and others (2008)^d Nakano and others (2006)

maximum identity of greater than 90 and less than 100 % mining was completed (Zhang and others 2012). As a result, 8,666 EST sequences were obtained. Redundant sequences with the same nucleotide were removed from the data set, and 6,800 final sequences were obtained. The EST annotation information of the obtained 6,800 sequences was used to analyze the expression patterns in different tissues.

RNA Isolation and Analysis of Quantitative Real-Time PCR in Melon

Total RNA was extracted using RNeasy Mini Kit (Qiagen) according to the manufacturer's instructions. Quantity and quality of RNA were checked by ND-2000 spectrophotometer. Total RNA was treated by DNase I to remove any genomic DNA contamination. First-strand cDNA was reverse transcribed from 2 mg of total RNA using an Omniscript kit (Qiagen) according to the manufacturer's instructions. BioRad CFX96 was used for quantitative real-time PCR (qRT-PCR) amplification using default parameters. The primer sequences are listed in Supplementary Table S2. The glyceraldehyde 3-phosphate dehydrogenase (GAPDH) gene was used as an internal standard to calculate relative fold differences based on the comparative Ct method. PCR reactions were carried out in 96-well optical reaction plates. The reaction was set up using 40 ng of cDNA sample as a template, 200 nM forward primers, 200 nM reverse primers, and SYBR green reaction mix. At least three biological and three technical replicates were performed for each reaction. The data were analyzed using the $\Delta\Delta C_t$ method. For calculating relative expression values during developmental stages, the leaf expression level was used as control. Data points in qRT-PCR time course are plotted as mean $\pm$ SE of three biological replicates.

Results

Identification of 136 AP2/ERF Superfamily Genes in Melon

To identify AP2/ERF superfamily genes in melon, BLASTp searches were performed using the MELO-NOMICS database with AP2/ERF domain of a melon ethylene response factor (GenBank number AB125975) as the query sequence. A total of 136 genes were identified as potentially containing the AP2/ERF domain (Table 1).

One hundred and nineteen genes that encoded proteins with a single AP2/ERF domain were assigned to the ERF family. The AP2 family, which could be identified based on a tandem repeat double AP2/ERF motif, was grouped into 13 genes. Four genes, containing a single AP2/ERF DNA-binding domain together with a B3 type domain, were assigned to the RAV family.

The same 136 sequences were obtained with the MELONOMICS database, by HMM analysis with PF00847, which contains a typical AP2/ERF domain. The individual ERF family genes are listed in Table S1.

Based on the amino acid identity of the AP2/ERF domains, the 119 ERF family members were classified into 2 subfamilies: 64 genes for ERF-like proteins were assigned to the ERF subfamily (Fig. S1), and the remaining 55 genes, which encode DREB-like proteins, constituted the DREB subfamily (Fig. S2). Overall, the 136 genes of the AP2/ERF superfamily in melon were divided into three families: ERF (119 genes), AP2 (13 genes), and RAV (4 genes), as shown in Table 1.

Phylogenetic Relationships Between ERF Family Members in Melon

To analyze their classifications and phylogenetic relationships, the amino acid sequences of the AP2/ERF domain in the

Fig. 1 Phylogenetic analysis of melon ERF family proteins. A radial neighbor-joining (Saitou and Nei 1987) phylogenetic tree was constructed using MEGA 5.10 (Tamura and others 2007) for the multiple sequence alignment of 119 melon ERF protein sequences. The 11 clades have been marked. The *dotted line* distinguishes between ERF and DREB subfamily proteins. *Scale bar* represents 0.05 amino acid substitutions per site

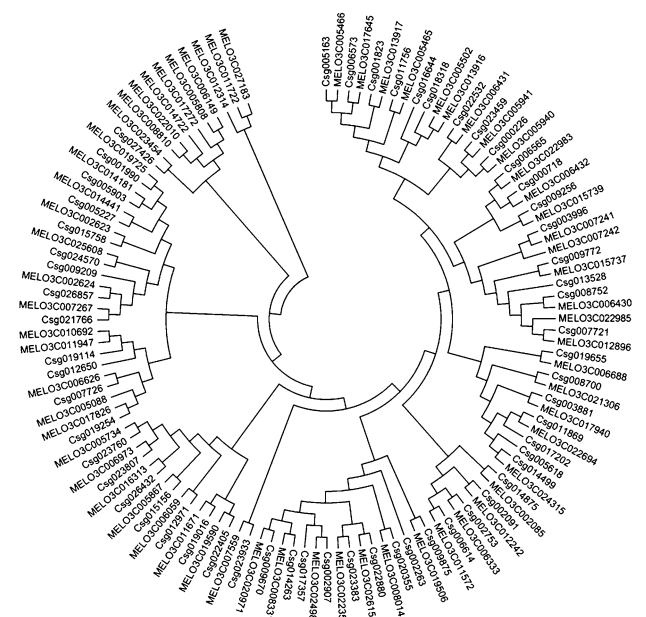
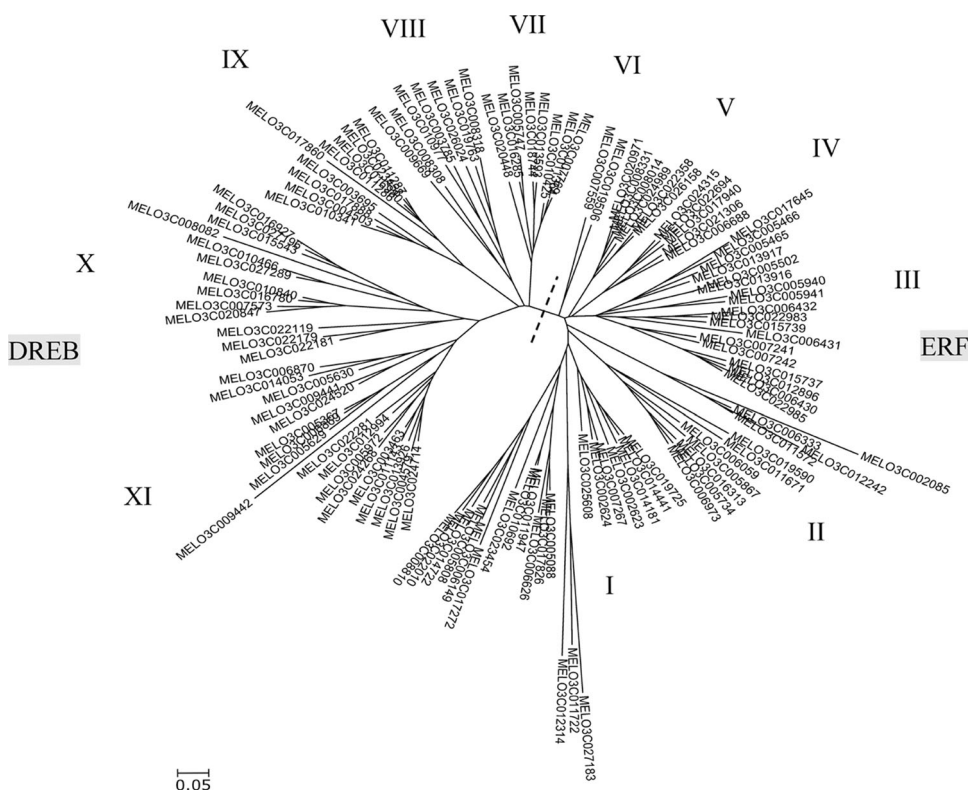


Fig. 2 Phylogenetic analysis of 64 melon and 58 cucumber ERF subfamily proteins. A phylogenetic tree by unrooted neighbor-joining (NJ) method (Saitou and Nei 1987) was constructed using MEGA 5.10 (Tamura and others 2007) for the multiple sequence alignment of 64 melon and 58 cucumber ERF subfamily protein sequences

119 melon ERF family proteins were analyzed by multiple alignment (Fig. S3). Residues Gly-4, Arg-6, Arg-8, Gly-11, Glu-17, Trp-38, Leu-39, Gly-40, Ala-48, Ala-49, and Gly-61

were conserved between all 119 proteins in melon (Fig. S3). In addition, greater than 95 % of ERF family members contained Val-5, Ile-18, Ala-19, Ala-51, Asp-53, and Ala-55 (Fig. S3). Based on this alignment, a radial NJ (Saitou and Nei 1987) phylogenetic tree was constructed by bootstrap analysis (1,000 replicates) using the multiple sequence alignments of the 119 ERF family proteins (Fig. 1).

Eleven distinct groups were identified in the phylogram and termed I to XI (Fig. 1; Table S1). Groups I to XI corresponded to groups that have been defined in *Arabidopsis* and rice (Nakano and others 2006). As shown in Fig. 1 and Table S1, ERF subfamily genes from melon were distributed between subgroups I, II, III, IV, V, and VI. The DREB subfamily genes were classified into 5 subgroups: VII, VIII, IX, X, and XI.

A phylogenetic tree by an unrooted NJ method (Saitou and Nei 1987) was constructed using MEGA 5.10 (Tamura and others 2007) for the multiple sequence alignment of 64 melon and 58 cucumber ERF subfamily protein sequences (Fig. 2). The resulting phylogenetic tree demonstrates that 10 ERF subfamily genes in melon and 1 subfamily gene in cucumber are thoroughly close in evolutionary relationship.

Conserved Motifs Outside of the AP2/ERF Domain

The conserved motifs in ERF family proteins in melon were analyzed using MEME Suite, version 4.8.1 (Figs. S5–15). The EAR motif was found only in proteins in

MELO3C019725	AI ^{black} PF ^{black} DL ^{black} NI ^{black} PP ^{black} PMD...VVDLTGDDLQATALCL	219
MELO3C014441	VI ^{black} PF ^{black} DL ^{black} NI ^{black} PP ^{black} PL ^{black} PP ^{black} HDV ^{black} DL ^{black} SGDDLHATALCF	214
MELO3C014181	PI ^{black} PF ^{black} DL ^{black} NI ^{black} P ^{black} SD...LFD ^{black} SGDDLRC ^{black} TALCL	277
MELO3C002623	GL ^{black} PF ^{black} DL ^{black} NE ^{black} PP ^{black} VWL.....	163
MELO3C007267	RI ^{black} DL ^{black} DL ^{black} NE ^{black} PP ^{black} ESA.....	246
MELO3C025608	PI ^{black} DE ^{black} DL ^{black} NI ^{black} PAVEVV.....	191
MELO3C002624	AI ^{black} DL ^{black} DL ^{black} NE ^{black} PP ^{black} DSA.....	211
Consensus	l dln p	

Fig. 3 Conserved EAR motif-like sequences in the C-terminal region of group I proteins in melon. *Black* and *gray* shades indicate identical and conserved amino acid residues in greater than 50 % of aligned sequences, respectively. Consensus amino acid residues are shown below the alignment

MELO3C019506	VFET ^{black} CSV ^{black} CQ ^{black} IEGCLGCN ^{black} FF ^{black} PP ^{black} SS ^{black} Q.....V	127
MELO3C007559	AMDE ^{black} CKK ^{black} CQ ^{black} FAGCLGCN ^{black} YF ^{black} EDGS ^{black} NQ ^{black} NQNVVKDEEND	109
Consensus	c cq gclgcn f s q	

Fig. 4 Conserved CX₂CX₄CX_{2~4}C sequences in the C-terminal region of group VI. *Black* and *gray* shades indicate identical and conserved amino acid residues in greater than 50 % of aligned sequences, respectively. Consensus amino acid residues are shown below the alignment

MELO3C010840	DLN ^{black} Q ^{black} FP ^{black} DP ^{black} ESS ^{black} DED.....	142
MELO3C016780	DLN ^{black} EY ^{black} PN ^{black} PET ^{black} S ^{black} DD.....	151
MELO3C007573	DLN ^{black} KL ^{black} PD ^{black} PEE ^{black} DEEGDGEFEWERVERH	173
MELO3C020847	DLN ^{black} K ^{black} FP ^{black} DP ^{black} ES ^{black} GD ^{black} CVSTHWTS.....	159
Consensus	dln p pe d	

Fig. 5 Conserved EAR motif-like sequences in the C-terminal region of group X proteins in melon. *Black* and *gray* shades indicate identical and conserved amino acid residues in greater than 50 % of aligned sequences, respectively. Consensus amino acid residues are shown below the alignment

groups I and X, as CMI-3 and CMX-4 motifs, respectively (Figs. 3, 4, 5).

The Cys repeat sequence CX₂CX₄CX_{2~4}C was observed only within the CMVI-2 motif in the N-terminal region of group X proteins (Fig. 4). Liu and others (1999) believe that regions that are rich in acidic amino acids, Gln, Pro, and/or Ser/Thr can usually be designated transcriptional activation domains. The conserved motifs that we identified have similar features, such as the abundance of Ser/Thr in group I as the CMI-4 motif (Fig. S5).

We also found that most motifs were distributed between different clades in the phylogenetic tree (Fig. 6), reflecting the structural similarities between members in the same group.

Structure and Evolution of ERF Family Genes

Most *Arabidopsis* ERF genes do not have introns (Sakuma and others 2002). Similarly, of the 119 ERF family genes that we identified, 87 (73.1 %) had no introns, and the remaining 32 had 1 or 2 introns in addition to CmERFI-22 and CmERFI-10, which have 6 and 8 introns, respectively (Table S1). The latter differs from the results in cucumber, which belongs to the same family (Hu and Liu 2011).

As shown in Table S1, 27 (84.4 %) of the 32 low-intron genes had a single intron, similar to the rate in cucumber (85 %), and in terms of numbers in melon the other 3 genes CmERFIV-5, CmERFIX-8, and CmERFII-1 possess 2. The number and position of the introns were conserved in each group, validating the reliability of our classification of melon ERF family genes (Hu and Liu 2011).

The genomic distribution of the 119 ERF family genes was analyzed to gain insight into their evolution. A similar situation also occurred among OsERF and AtERF members (Nakano and others 2006), indicating that ERF genes are distributed widely throughout the genome of the common ancestor of monocots and eudicots.

Expression Analysis of 64 Melon ERF Subfamily Genes

The expressed sequence tags numbered 131,052 from *Cucumis melo* have been deposited in GenBank (<http://www.ncbi.nlm.nih.gov/nucest/?term=melon>). These partial transcript sequences forms, which are valuable for gene expression pattern observation of the expressional characteristics and abundance of genes, are revealed via these partial transcript sequences. To demonstrate the patterns of

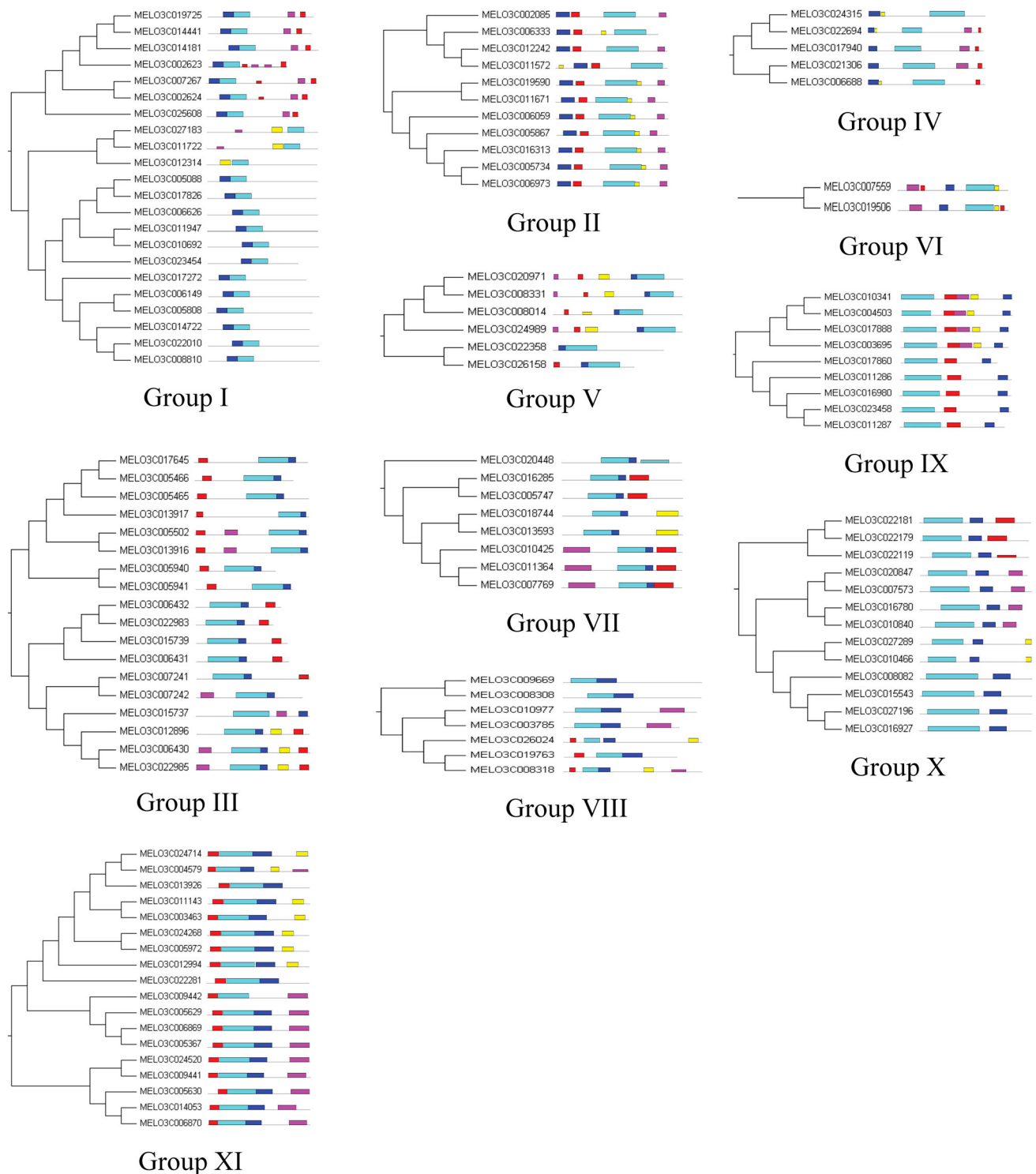


Fig. 6 Phylogenetic relationships between melon ERF family genes. Bootstrap values from 1000 replicates were used to assess the robustness of the trees. A phylogenetic tree and schematic of the protein structures of each group are shown. Aqua Blue color represents AP2/ERF domain in all groups. In group I and X, red and pink color represent EAR motif; in group VI, blue color represent

CX₂CX₄CX_{2~4}C motif, respectively. In other groups, conserved motifs which are different from EAR/CX₂CX₄CX_{2~4}C motifs in different groups are summarized with the same colors, respectively, in Supplementary Figs. S5–15. These motifs were defined by multiple alignment with MEME Suite version 4.8.1 (Color figure online)

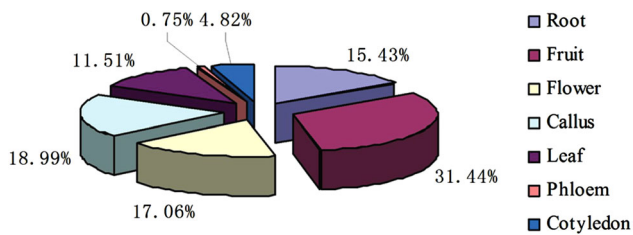


Fig. 7 Distribution of ERF subfamily genes in different melon tissues or organs in silico. The *number* indicates the percentage of ESTs that are detected in melon tissue/organ type, regardless of the expression level

the expression of these ERF subfamily genes across various tissues, an analysis of gene expression in seven tissues was performed. Patterns of expression of ERF subfamily genes in melon were determined by analyzing the EST counts based on total EST sequences (8,666 EST sequences) which had a maximum identity of greater than 90 and less than 100 % in BLASTn screening in the NCBI EST database. The ERF subfamily genes were detected in 7 kinds of tissue (fruit, flower, callus, leaf, root, cotyledon, and phloem) (Fig. 7). Among them, fruit had the highest expression level of 31.44 % with callus in second place at 18.99 %. This was followed by 17.06, 15.43, 11.51, 4.82, and 0.75 % for flower, root, leaf, cotyledon, and phloem, respectively. The percentage of ERF subfamily genes expressed in phloem (0.75 %) was the lowest in all tissues (Fig. 7). To further study these ERF subfamily genes expression pattern of the single gene among transcriptomes of different tissues, expression profiles of 64 ERF subfamily genes of melon which is in group I–VI were examined by analyzing EST counts (Figs. 8, 9, 10, 11). In this study, redundant sequences with the same nucleotide were removed from the data set, and 6,800 final sequences were obtained, which were used to observe gene expression pattern in different tissues. We observed that CmERFIII-11 tends to present its specific expression in melon fruit and

callus with multiple copy forms (Fig. 10). CmERFII-9 was expressed in flower which was highest in all tissues (Fig. 9). Four genes (CmERFII-9, CmERFIV-3, CmERFV-2, and CmERFVI-2) were randomly selected from three tissues (leaf, root, and fruit) for validation of expression patterns of melon ERF subfamily genes using qRT-PCR analysis (Fig. 12). Consistent with the expression patterns of melon ERF subfamily genes, the resulting of qRT-PCR analysis supports the reliability of expression patterns.

Discussion

Nakano and others (2006) systematically surveyed the gene structures, phylogeny, and conserved motifs of the ERF gene family in *Arabidopsis* and rice, but few melon ERF genes have been studied. In this report, we performed a comprehensive analysis of genes that encode AP2/ERF proteins in the melon genome, identifying 136 AP2/ERF superfamily genes from the MELONOMICS database.

To examine the phylogenetic relationships between melon ERF family proteins, we performed multiple alignments using the amino acid sequences of the AP2/ERF domains. All ERF subfamily genes that contained the WLG motif (W27, L28, and G29), except CmERFI-12, CmERFI-10, and CmERFI-22, harbored Y27 (Fig. S1), and all DREB subfamily genes with a WLG motif (W27, L28, and G29), except CmERFXI-10, contained V28 (Fig. S2). Based on the alignments of ERF family proteins with the full-length amino acid sequences from melon, there was more variation among those outside of the AP2/ERF domain, as in *Populus trichocarpa* (Zhuang and others 2008).

The bootstrapping values for the nodes in this phylogenetic tree were not high for every case, similar to an earlier phylogenetic analysis of *Arabidopsis* ERF proteins (Nakano and others 2006), likely due to the little variation in AP2/ERF domains and the highly conserved members

Fig. 8 Expression patterns of melon ERF subfamily genes in group I in various tissues. The expression profile was determined by analyzing the EST counts based on non-redundant EST sequences of melon. The unit of the y-axis is the number of ESTs

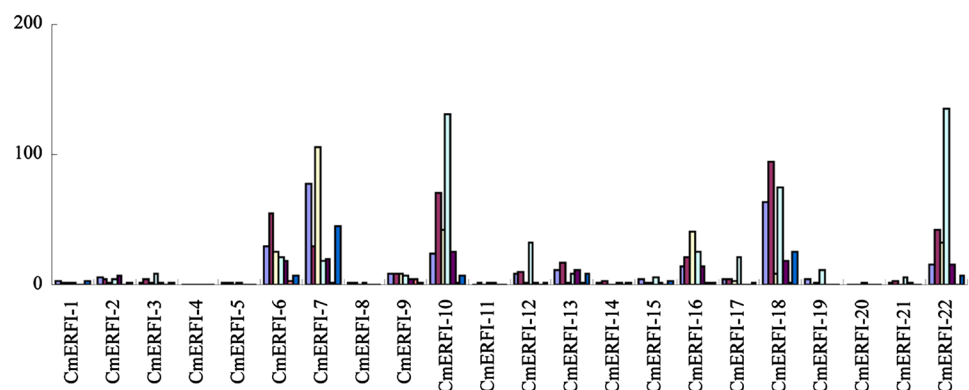


Fig. 9 Expression patterns of melon ERF subfamily genes in group II in various tissues. The expression profile was determined by analyzing the EST counts based on non-redundant EST sequences of melon. The unit of the y-axis is the number of ESTs

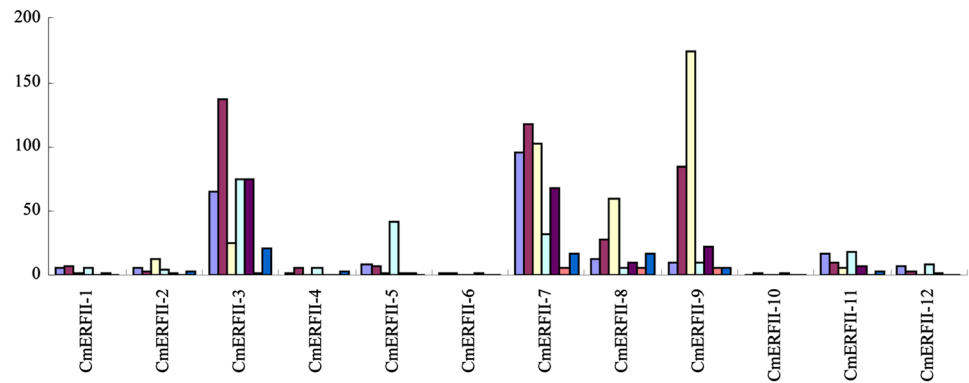


Fig. 10 Expression patterns of melon ERF subfamily genes in group III in various tissues. The expression profile was determined by analyzing the EST counts based on non-redundant EST sequences of melon. The unit of the y-axis is the number of ESTs

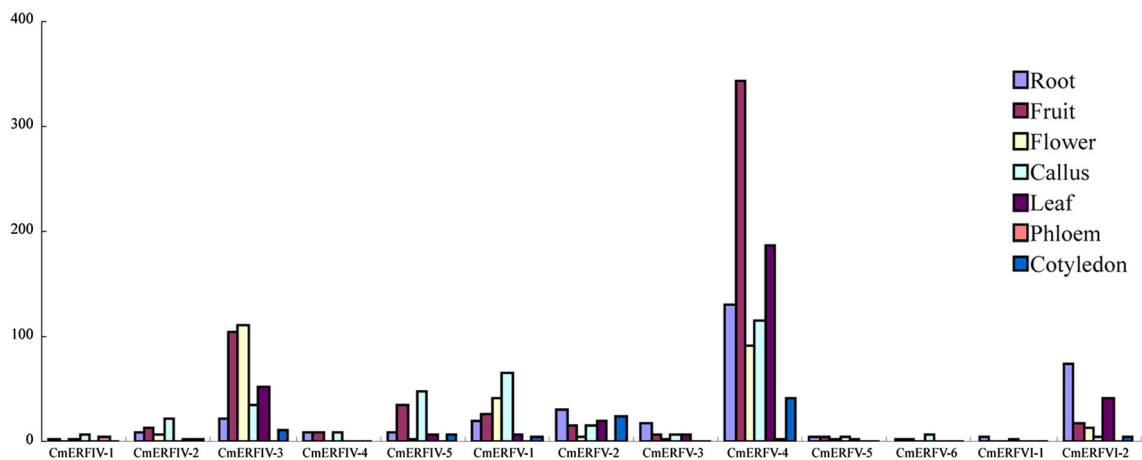
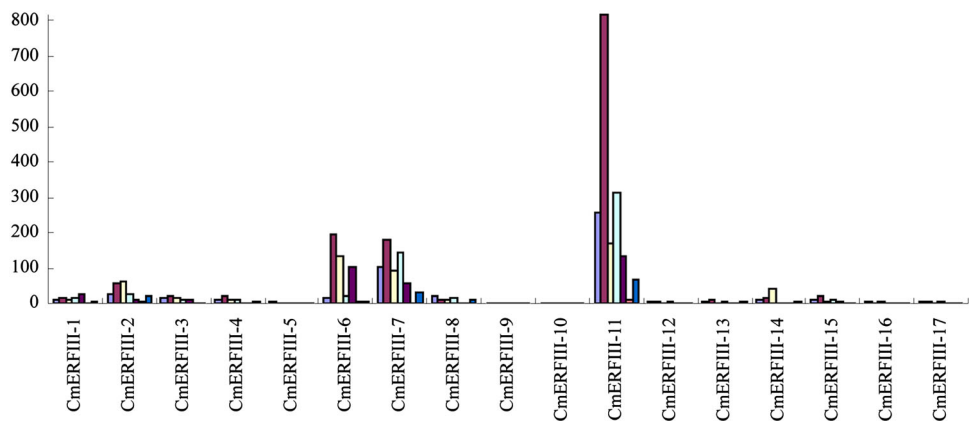


Fig. 11 Expression patterns of melon ERF subfamily genes in group IV, V, and VI in various tissues. The expression profile was determined by analyzing the EST counts based on non-redundant EST sequences of melon. The unit of the y-axis is the number of ESTs

within the subfamily. Another phylogenetic tree was constructed using the maximum parsimony (MP) method (Fig. S4) to verify the reliability of the NJ method-based tree, in which nearly all ERF family members were placed in the same groups. The resulting phylogenetic tree demonstrates that 10 ERF subfamily genes in melon are different from 1 gene in cucumber. The possible reason is due to the close phylogeny, genes which are classified in different species

lead to gene function differentiation, such as shape, sweetness, texture, and so on.

In transcription factors, regions that lie outside of the DNA-binding domain contain functionally important motifs that mediate transcriptional activity, protein–protein interactions, and nuclear localization (Liu and others 1999). Proteins of a group that shares these domains or motifs in a phylogenetic tree are likely to have similar

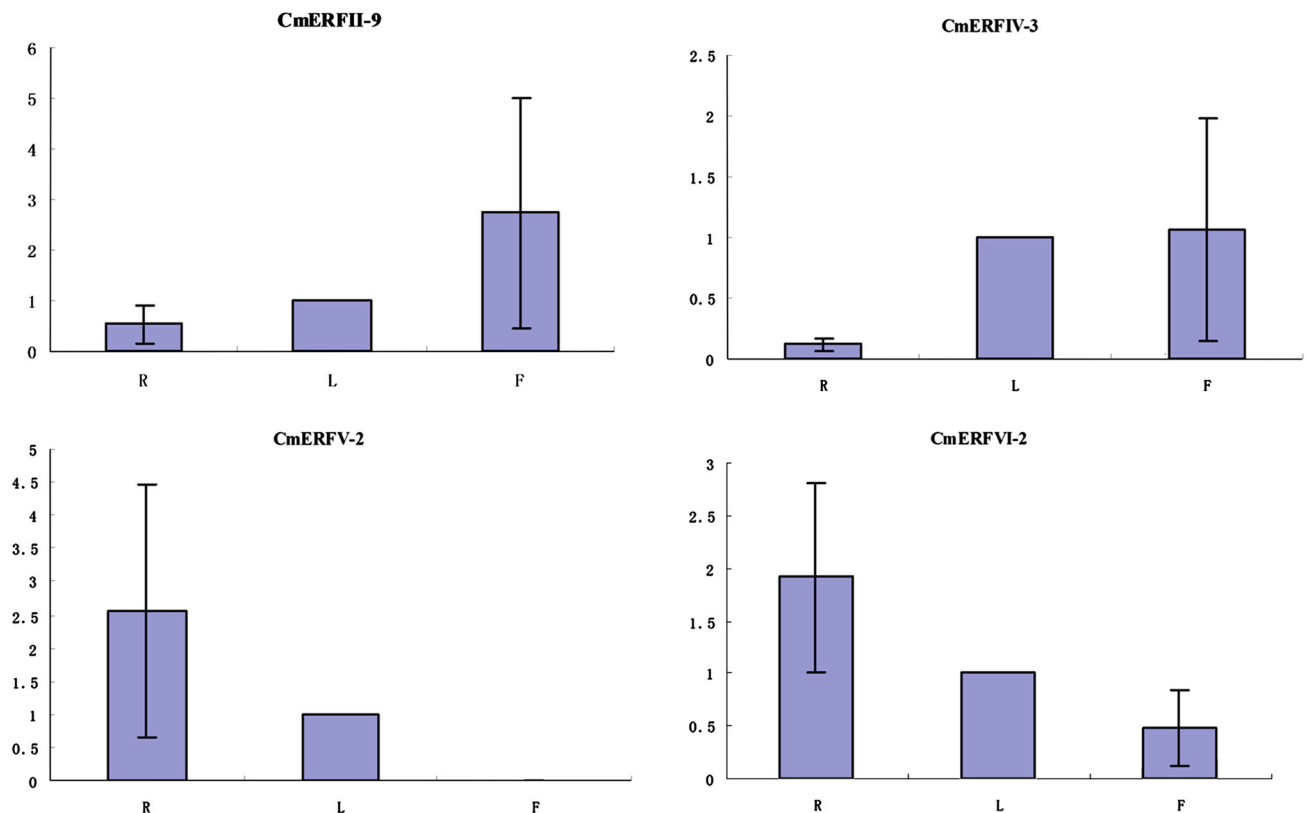


Fig. 12 Validation of expression patterns of melon ERF subfamily genes using qRT-PCR analysis. The expression data obtained using the EST counts based on non-redundant EST sequences for selected 4 genes were validated using qRT-PCR analysis. The results were expressed using the leaf as a reference for each gene (relative mRNA

level 1). Values represent the best experiment among three independent biological repetitions. y-axis represents relative expression values, and X-axis represents stages of development as follows: *L* leaf, *R* roots, and *F* fruit. Bars indicate the SEM of three experimental repetitions

functions. The ERF-associated amphiphilic repression (EAR) motif (DLNxxP), a repression domain in the C-terminal regions of repressor-type ERF proteins, mediates several functions by negatively regulating genes that participate in developmental, hormonal, and stress signaling pathways (Fujimoto and others 2000; Ohta and others 2001). In our motif analysis, the EAR motif existed only in proteins in groups I and X, such as CMI-3 and CMX-4, (Figs. 3, 5). Thus, we speculate that these genes are involved in negative regulation.

The Cys repeat sequence CX₂CX₄CX_{2~4}C, possibly a zinc-finger motif, participates in DNA-binding or protein-protein interactions (Nakano and others 2006). This consensus sequence was only found in the CMVI-2 motif of the N-terminal region of group X proteins. Liu and others (1999) believed that regions that are rich in acidic amino acids, Gln, Pro, and/or Ser/Thr can usually be designated transcriptional activation domains. The conserved motifs that we identified shared features, such as an abundance of Ser/Thr in Group I, as noted for the CMI-4 motif (Figs. 4, S5).

Because gene expression patterns often correlate with a gene's function, ESTs, created by partially sequencing randomly isolated gene transcripts, have been invaluable in discovery through expression pattern analysis. Identification of ERF subfamily genes and the analysis of gene expression patterns through large amounts of ESTs are classic examples. In this study, to detect the expression patterns of the 64 genes in the ERF subfamily of melon, seven tissue types were analyzed according to the annotation of ESTs (Fig. 7). Transcripts of the genes were the most abundant in fruit tissue, followed by callus and flower tissue. Most of the ERF subfamily genes were detected in all seven types of tissues. The CmERFIII-11 gene was particularly interesting in that its level of detection was the highest among all ERF subfamily genes, and was reached in fruit tissue. Therefore, we speculate that the CmERFIII-11 gene may play an important role mainly in fruit growth and development in melon. In addition, CmERFIII-11 with high abundant ESTs has been detected in callus, suggesting a function for this gene in stress response. Some genes, such as CmERFIII-10 and CmERFI-4, failed to match in

ESTs, because they might lack ERF activity as pseudo-genes or because the data in the EST database were insufficient to cover the expression of all genes. The natural gene expression levels in various tissues were not incarnated via the frequencies of EST occurrence present in GenBank. However, it is reasonable for researchers that they obtain the outline for the gene expression pattern with their interest. This study would be of interest to plant molecular biologists, particularly those interested in the ERF subfamily genes. They could use the work included in this study to predict the proteins in other types of plants.

Conclusion

After extensive analysis, 119 ERF family genes were identified and divided into 11 groups, I to XI, consistent with previous studies (Sharma and others 2010). This classification was based on the presence and position of the introns and the conserved motifs outside of the AP2/ERF domain. Analysis of expression patterns shows that ERF subfamily genes were distributed widely throughout melon plant tissues, and most of the ERF subfamily genes were detected in all seven types of tissue. Some ERF subfamily genes show expression patterns characteristic of tissue-specific expression.

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